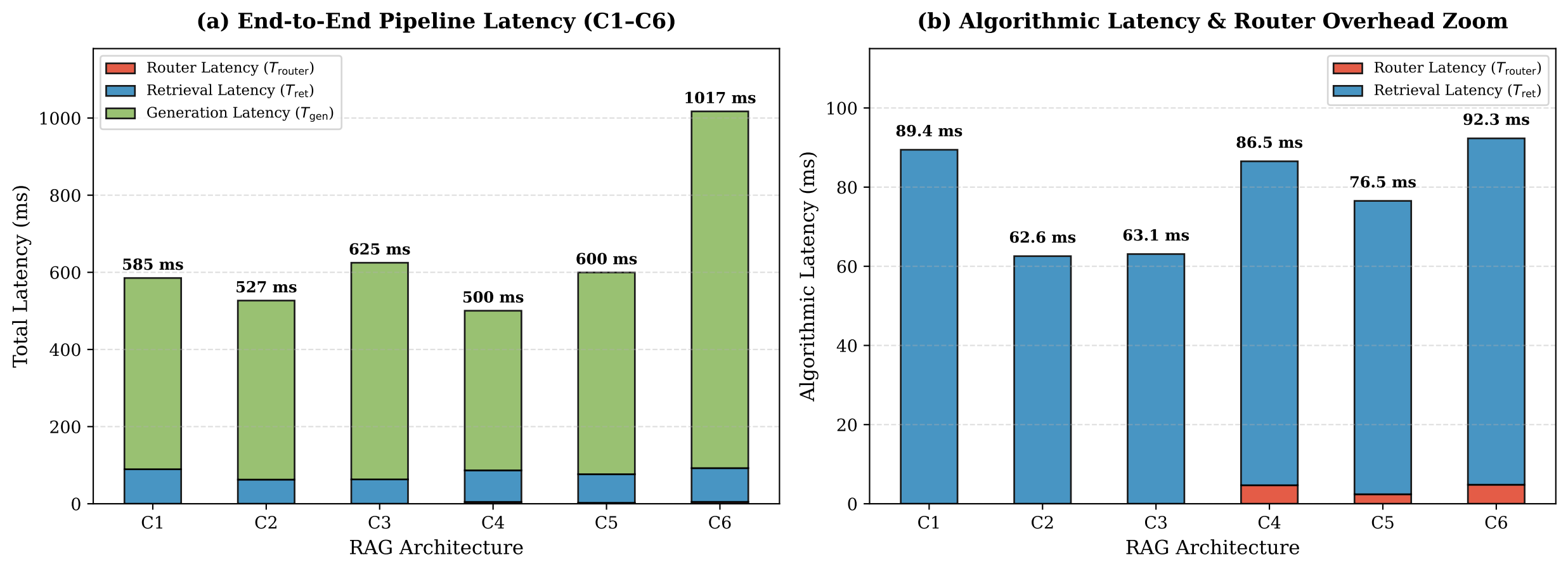
(a) End-to-End Pipeline Latency (C1–C6)
Router Latency (T_{router})
Retrieval Latency (T_{ret})
Generation Latency (T_{gen})
Total Latency (ms)
0
200
400
600
800
1000
585 ms
527 ms
625 ms
500 ms
600 ms
1017 ms
C1
C2
C3
C4
C5
C6
RAG Architecture
(b) Algorithmic Latency & Router Overhead Zoom
Router Latency (T_{router})
Retrieval Latency (T_{ret})
Algorithmic Latency (ms)
0
20
40
60
80
100
89.4 ms
62.6 ms
63.1 ms
86.5 ms
76.5 ms
92.3 ms
C1
C2
C3
C4
C5
C6
RAG Architecture